# EXCELLENCE

## CLASS 11 · 2026–27

Physics | Lecture - 04

# Basic Maths and Calculus

**Leftover last class, then ASTC, max–min, phase**

By — Excellence Programme

4 September 2026

# Topics to be covered

**01** Full table 0° to 360° for all six ratios, then arc and radian ↔ degree

**02** Last-class homework — slides 21, 23, 25, 33, 35, 37, 49 — answers in simple lines

**03** Length prefixes with full names, then compound sin / cos / tan

**04** Pending last class, slides 71–82 — small angle, then allied homework

**05** ASTC method — sign first, then change the function only at 90° and 270°

**06** Maximum and minimum, then phase / phasor

# All six ratios from 0° to 90°

| θ | 0° | 30° | 45° | 60° | 90° |
|---|---|---|---|---|---|
| **sin** | 0 | 1/2 | 1/√2 | √3/2 | 1 |
| **cos** | 1 | √3/2 | 1/√2 | 1/2 | 0 |
| **tan** | 0 | 1/√3 | 1 | √3 | undef. |
| **cot** | undef. | √3 | 1 | 1/√3 | 0 |
| **sec** | 1 | 2/√3 | √2 | 2 | undef. |
| **cosec** | undef. | 2 | √2 | 2/√3 | 1 |

tan 90° and sec 90° are not defined. cosec 0° and cot 0° are not defined.

# All six ratios from 90° to 180°

| θ | 90° | 120° | 135° | 150° | 180° |
|---|---|---|---|---|---|
| **sin** | 1 | √3/2 | 1/√2 | 1/2 | 0 |
| **cos** | 0 | −1/2 | −1/√2 | −√3/2 | −1 |
| **tan** | undef. | −√3 | −1 | −1/√3 | 0 |
| **cot** | 0 | −1/√3 | −1 | −√3 | undef. |
| **sec** | undef. | −2 | −√2 | −2/√3 | −1 |
| **cosec** | 1 | 2/√3 | √2 | 2 | undef. |

Second quadrant: only sin and cosec are positive. cot 180° and cosec 180° are not defined.

# All six ratios from 180° to 270°

| θ | 180° | 210° | 225° | 240° | 270° |
|---|---|---|---|---|---|
| sin | 0 | −1/2 | −1/√2 | −√3/2 | −1 |
| cos | −1 | −√3/2 | −1/√2 | −1/2 | 0 |
| tan | 0 | 1/√3 | 1 | √3 | undef. |
| cot | undef. | √3 | 1 | 1/√3 | 0 |
| sec | −1 | −2/√3 | −√2 | −2 | undef. |
| cosec | undef. | −2 | −√2 | −2/√3 | −1 |

Third quadrant: only tan and cot are positive. tan 270° and sec 270° are not defined.

# All six ratios from 270° to 360°

| θ | 270° | 300° | 315° | 330° | 360° |
|---|---|---|---|---|---|
| sin | −1 | −√3/2 | −1/√2 | −1/2 | 0 |
| cos | 0 | 1/2 | 1/√2 | √3/2 | 1 |
| tan | undef. | −√3 | −1 | −1/√3 | 0 |
| cot | 0 | −1/√3 | −1 | −√3 | undef. |
| sec | undef. | 2 | √2 | 2/√3 | 1 |
| cosec | −1 | −2/√3 | −√2 | −2 | undef. |

Fourth quadrant: only cos and sec are positive. 360° repeats 0°. Never write cosec 360°.

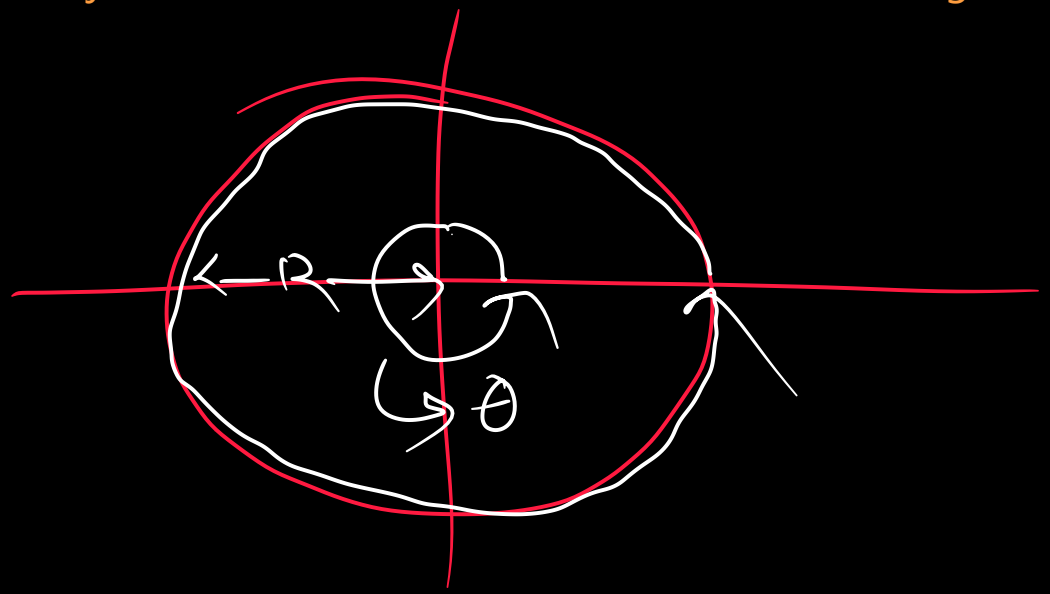
# Arc length — θ must be in radian

## Arc length = R θ

**θ in radian**

Practical unit of angle: degree. SI unit of angle: radian.
+θ anti-clockwise. −θ clockwise.

If the object turns 90° on a circle of radius R, first change 90° to π/2. Distance = πR/2. Never R × 90°.




$\theta = \frac{360}{} = 2\pi$ Radian

$R\theta = R2\pi \rightarrow 2\pi R$

↳ = circumference of circle

If $\theta = 120° \Rightarrow$ arc length = ? $\Rightarrow$

# Rough work / extra work

$\theta = 120^\circ \Rightarrow \rightarrow \underline{120^\circ} \Rightarrow$ Radian

↳ 1 degree $\Rightarrow \frac{\pi}{180}$

$1 \text{ Rad} = \frac{180}{\pi}$

$1 \text{ Rad} = \frac{180^\circ}{\pi}$

$\frac{120\pi}{180} = \frac{2}{3}\pi \cdot R$

$\text{arc length} = \frac{2}{3}\pi R$

## The one line: 180° = π rad

VIP

Lock this. Every conversion starts here.

**180° = π rad**

**Wherever you see π rad, put 180°.**

**degree → radian: multiply by π/180**
**radian → degree: multiply by 180/π**

Never write π = 180°. Write π rad = 180°. π with no degree mark is already radian.

# Homework slide 21 — six conversions

**Last class, slide 21. Convert each. Easy rule: π rad = 180°.**

If you see π, put 180°. If you see a degree, multiply by π/180.

**π/2 rad = 180°/2 = 90°**

**π/6 rad = 180°/6 = 30°**

**90° = 90 × π/180 = π/2 rad**

**π/4 rad = 180°/4 = 45°**

**45° = 45 × π/180 = π/4 rad**

**60° = 60 × π/180 = π/3 rad**

# Homework slide 23 — radian into degree

**Last class, slide 23. Example: 5π/6 means 5 × 180° / 6.**

| | | |
|---|---|---|
| 5π/6 = 150° | 2π/3 = 120° | 4π/3 = 240° |
| 3π/5 = 108° | 5π = 900° | 2π/5 = 72° |
| 7π/12 = 105° | 5π/2 = 450° | 3π = 540° |

How: cancel. 5π/6 → 5 × 180 / 6 → 5 × 30 = 150. Same for every line.

$\frac{\pi}{3} = 60 \times 4 = 240$

$= 180 \times 3 = 540$

$5\left(\frac{\pi}{6}\right) \Rightarrow 5 \times 30 \Rightarrow 150$

$\frac{108}{5} \rightarrow \frac{3 \times 180}{5} \Rightarrow 36 \times 3 = 108$

# Rough work / extra work

$5\pi = 900 \quad 5 \times 180 = 900$

$\frac{2\pi}{3} = 2 \times \frac{180}{3} = 2 \times 60 = 120$ (60)

$\frac{2\pi}{5} = ? \Rightarrow 2 \times \frac{36\pi}{5} \Rightarrow 2 \times 36 = 72$

# Homework slide 25 — degree into radian

HW send solution

**Last class, slide 25. Multiply the degree by π/180. Cancel 180 if you can.**

**120° = 2π/3 rad**

**30° = π/6 rad**

**150° = 5π/6 rad**

**270° = 3π/2 rad**

**135° = 3π/4 rad**

**π is already radian**

Wrong: π° treated as π rad

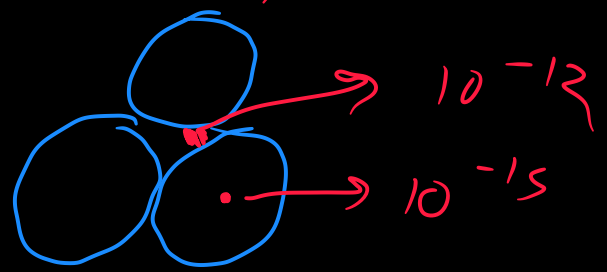
Right: The board wrote π, not π°. π is already radian. π° would be $\pi^2/180$ rad.

# Tiny lengths you will meet in physics

A prefix is a short name stuck in front of metre. It tells you how many metres.

| short | full name | in metres | where you meet it |
|---|---|---|---|
| **fm** | femtometre | $10^{-15}$ m | nucleus of an atom |
| **pm** | picometre | $10^{-12}$ m | spacing inside an atom |
| **Å** | ångström | $10^{-10}$ m | atom size, X-ray wavelength |
| **nm** | nanometre | $10^{-9}$ m | visible light, DNA, thin films |
| **µm** | micrometre | $10^{-6}$ m | cell, dust, width of a hair |

Å is not SI, but physics still uses it for atoms. Nuclear size is fm, never mm.



# Everyday and large lengths

| short | full name | in metres | where you meet it |
|---|---|---|---|
| **mm** | millimetre | $10^{-3}$ m | small lab length, thickness |
| **cm** | centimetre | $10^{-2}$ m | ruler, height in class |
| **m** | metre | 1 m | SI unit of length |
| **km** | kilometre | $10^{3}$ m | road, tower, walking distance |
| **Mm** | megametre | $10^{6}$ m | Earth radius ≈ 6.4 Mm |

**1 m = 100 cm** **1 km = 1000 m** **1 Å = $10^{-10}$ m** **1 fm = $10^{-15}$ m**

# How to read a prefix in one line

**milli = one thousandth** **kilo = one thousand**

**micro = one millionth** **nano = one billionth**

**pico = one trillionth** **femto = one thousand-trillionth**

1000 = 1k

3000 = 3k

1 mm is 1 m cut into 1000 equal pieces. 1 km is 1000 metres joined.

1 µm is 1 m cut into 1 000 000 pieces. A red-blood-cell is a few µm.

Never mix them: a nucleus is a few fm. An atom is about 1 Å. Light is a few hundred nm.

# Compound sine — two angles, one sine

A and B are two angles. Compound means we open the bracket into products.

**sin (A + B) = sin A cos B + cos A sin B**

In words: sine of a sum = sine-cosine plus cosine-sine.

**sin (A − B) = sin A cos B − cos A sin B**

Same line, only the middle sign flips. Difference takes a minus.

Do not write sin(A+B) = sin A + sin B. That is the classic trap.

H.W

Short notes

# Compound cosine — watch the minus

**cos (A + B) = cos A cos B − sin A sin B**
In words: cosine of a sum = cosine-cosine minus sine-sine.

**cos (A − B) = cos A cos B + sin A sin B**
Difference of cosine takes a plus. Sum of cosine takes a minus. Opposite of sine.

Swap those two signs and every 15°, 75°, 105° value will come out wrong.

M.D

Short notes

$\Rightarrow \sin(A+B) = \sin A \cdot \cos B + \sin B \cdot \cos A$

$\sin(A-B) = \sin A \cdot \cos B - \sin B \cos A$

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos(A-B) = \cos A \cos B + \sin A \cdot \sin B$

# Compound tangent — one fraction

**tan (A + B) = (tan A + tan B) / (1 − tan A tan B)**

Top: add the tans. Bottom: 1 minus the product. Sum has a minus downstairs.

**tan (A − B) = (tan A − tan B) / (1 + tan A tan B)**

Difference: minus on the top, plus downstairs.

**If tan A tan B = 1, then tan(A + B) is not defined.**

$$\frac{\tan(A) + \tan B}{1 - \tan A \cdot \tan B}$$

# When A and B are the same — double angle

Put B = A in the compound lines. Two of the same angle become 2A.

**$\sin 2A = 2 \sin A \cos A$**

**$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$**

**$\tan 2A = 2 \tan A / (1 - \tan^2 A)$**

sin 2θ is not 2 sin θ. The 2 sits with the angle, not in front of the sine.

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$ or $1 - 2\sin^2 A$ or $2\cos^2 A - 1$

# Rough work / extra work

$\sin(A+B) = \sin A\cos B + \sin B\cos A$

$= \sin(A + A)$ ⇒ ✗

$B = A$

$\sin A\cos(A) + \sin(A)\cos(A)$

$= 2\times\sin A\cos A$

## Rough work / extra work

$\cos(2A) = \cos(A+A) \Rightarrow$

$\cos(A+B) = \cos A \cdot \cos B - \sin A \sin B$

$B = A$

$\cos(A+A) = \cos(A) \cdot \cos(A) - \sin(A) \cdot \sin(A)$

$= \cos^2 A - \sin^2 A$

$\sin^2 A$

$= 1 - \cos^2 A$

$\cos(2A) = \cos^2 A - (1 - \cos^2 A)$

# Rough work / extra work

$$\cos(2A) = \cos^2 A - 1 + \cos^2 A$$

$$= 2\cos^2 A - 1 \quad \text{— (1)}$$

$$\sin^2 A + \cos^2 A = 1 \Rightarrow \cos^2 A = 1 - \sin^2 A \quad \text{— (2)}$$

Put value of $\cos^2 A$ into Eq (1)

$$= 2(1 - \sin^2 A) - 1$$

$$2 - 2\sin^2 A - 1$$

$$1 - 2\sin^2 A = \cos(2A)$$

# Homework slide 33 — shadow of a pole

**Last class, slide 33. Pole height h. Shadow l. Sun at angle θ above the ground. Find l.**

1. l = h tan θ
2. l = h / tan θ
3. l = h sin θ
4. l = h / sin θ

Picture: height is the wall, shadow is the floor. tan θ = wall / floor = h / l.

**So l = h / tan θ (same as h cot θ)**

**Option 2 l = h / tan θ**

Option 1 swaps height and shadow. tan is not shadow/height.

# Homework slide 35 — wire falls

H.W

**Last class, slide 35. Wire from the top at angle θ breaks and falls. Find x from the base.**

1. x = h cot θ
2. x = h tan θ
3. x = h sin θ
4. x = h sec θ

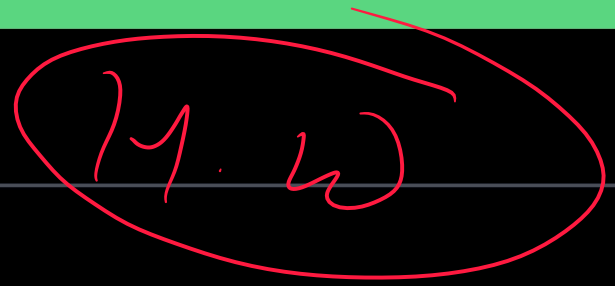
Wire is the long side = h / sin θ. Original base = h / tan θ. After the fall, x is wire minus base.

**x = h / sin θ − h / tan θ = h (cosec θ − cot θ)**

**x = h (cosec θ − cot θ) — none of the four options**

Option 1 is only the original base, before the wire falls.

# Homework slide 37 — shadow on a slope



**Last class, slide 37. Pole 10 m. Slope 30°. Sun 60°. Shadow length l on the slope?**

1. 5 m
2. 10 m
3. 10√3 m
4. 10 / (√3 − 1) m

Rise of the slope over length l is l/2. Horizontal run is l√3/2. tan 60° looks at (10 + rise) over run.

**√3 = (10 + l/2) / (l √3 / 2)  3l/2 = 10 + l/2  l = 10**

**Option 2  l = 10 m**

# Homework slide 49 — four values, easy route

**Last class, slide 49. Find sin 120°, cos 150°, sin 240°, cos 120°.**

Easy way: write each as 180° with a small angle. ASTC then gives the sign.

**(1) sin 120° = sin (180° − 60°) = + sin 60° = √3/2**

**(2) cos 150° = cos (180° − 30°) = − cos 30° = −√3/2**

**(3) sin 240° = sin (180° + 60°) = − sin 60° = −√3/2**

**(4) cos 120° = cos (180° − 60°) = − cos 60° = −1/2**

**√3/2 −√3/2 −√3/2 −1/2**

# Last class stopped at slide 71 — small angle now

Last class ended before small angle. Slides 71 to 82 of that deck are today’s leftover.

**small-angle rule → sin 2°, tan 3°, cos 3° → tan 2° trap → allied homework**

Then we start the new work: ASTC method, max–min, phase.

# When θ is small, the arc looks like the opposite side

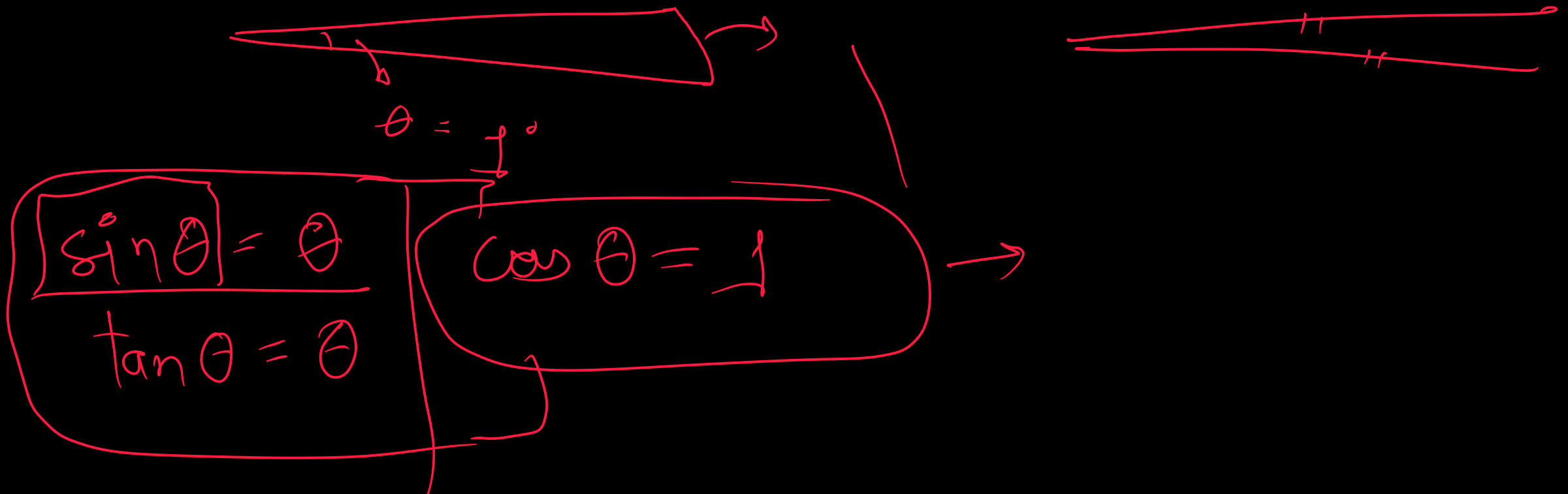
On a circle of radius R, a tiny angle makes opposite ≈ arc = Rθ, and adjacent ≈ R.

**sin θ ≈ θ** **tan θ ≈ θ** **cos θ ≈ 1**

**θ in radian. Use only if θ ≤ 5°.**

Easy words: change the degree to radian first, then drop the word radian. That number is the answer.

sin 2° is not 2. 2° in radian is 2π/180 = π/90. That is the value.



# When θ is small, the arc looks like the opposite side

On a circle of radius R, a tiny angle makes opposite ≈ arc = Rθ, and adjacent ≈ R.

**sin θ ≈ θ** **tan θ ≈ θ** **cos θ ≈ 1**

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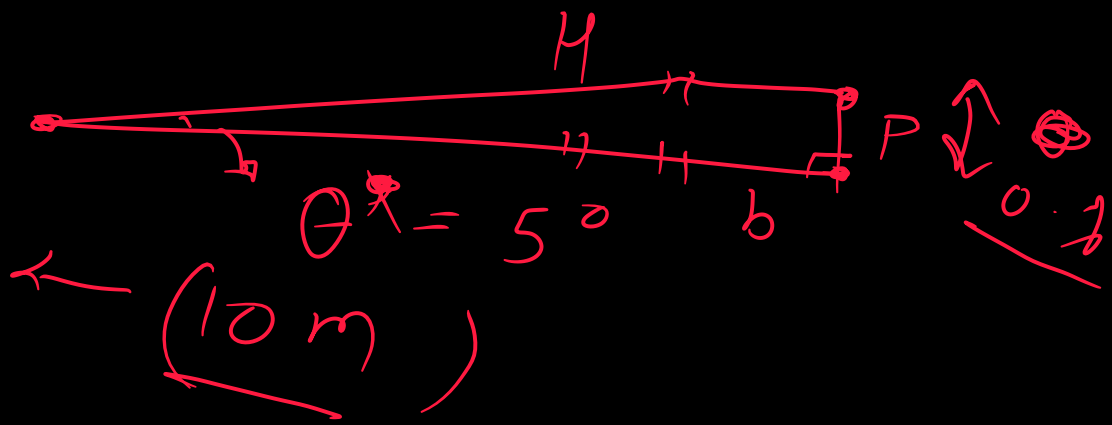
sin 2° is not 2. 2° in radian is 2π/180 = π/90. That is the value.

$\theta = 5^\circ$

$H^2 = P^2 + b^2$

$H^2 = 0.01 + 100$

$H^2 = 100.01 \approx 100$



# Pending slide 72 — sin 2°, tan 3°, cos 3°

**Last class, slide 72. Using small-angle, find sin 2°, tan 3°, cos 3°.**

Step 1: the angle is small, so sine and tan become the radian number. Cosine becomes 1.

**(i) sin 2° ≈ 2 × (π/180) = π/90**

**(ii) tan 3° ≈ 3 × (π/180) = π/60**

**(iii) cos 3° ≈ 1**

**sin 2° ≈ π/90 tan 3° ≈ π/60 cos 3° ≈ 1**

$\sin 2^\circ = 2^\circ \Rightarrow \frac{2 \times \pi}{180} \Rightarrow \frac{\pi}{90}$

$\tan 3^\circ = ? = 3^\circ \Rightarrow \frac{3 \times \pi}{180} = \frac{\pi}{60}$

$\cos 3^\circ = 1$

# Rough work / extra work

# Pending slide 74 — which tan 2° is correct?

**Last class, slide 74. Which is correct for tan 2° under small-angle?**

1. 2°
2. ~~π / 90~~
3. π / 30
4. 2 rad

$\tan 2^\circ = 2^\circ = 2 \times \frac{\pi}{180} = \frac{\pi}{90}$

Wrong: tan 2° = 2°

Right: tan 2° ≈ 2 × (π/180) = π/90

**Option 2 tan 2° ≈ π/90**

Option 1 leaves the degree. Option 4 pretends 2° is already 2 radian. Both are wrong.

# Pending slides 76–77 — two allied lines

**Last class, slides 76–77. Simplify. (13) $\sin(180^\circ - \theta)$ (14) $\cos(180^\circ + \theta)$**

$\sin(180^\circ - \theta) = 0 \cdot \cos\theta - (-1)\sin\theta = \sin\theta$

**(13) $\sin(180^\circ - \theta) = \sin\theta$**

$\cos(180^\circ + \theta) = (-1)\cos\theta - 0 = -\cos\theta$

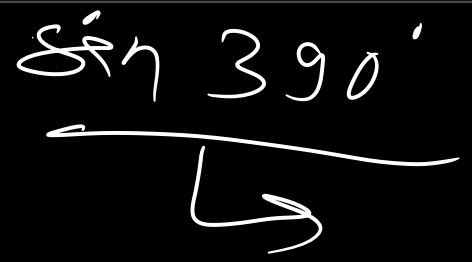
**(14) $\cos(180^\circ + \theta) = -\cos\theta$**

# Pending slide 77 — match the column

**Last class, slide 77. Match. Modulus means size, ignore the minus.**

| Column I | easy working | match |
|---|---|---|
| **(A) sin 390°** | same as sin 30° = +1/2 | P and T |
| **(B) sin (−30°)** | minus, so −1/2 | Q and T |
| **(C) cos 120°** | minus, so −1/2 | Q and T |
| **(D) tan (−120°)** | comes out +√3 which is bigger than 1 | P and S |

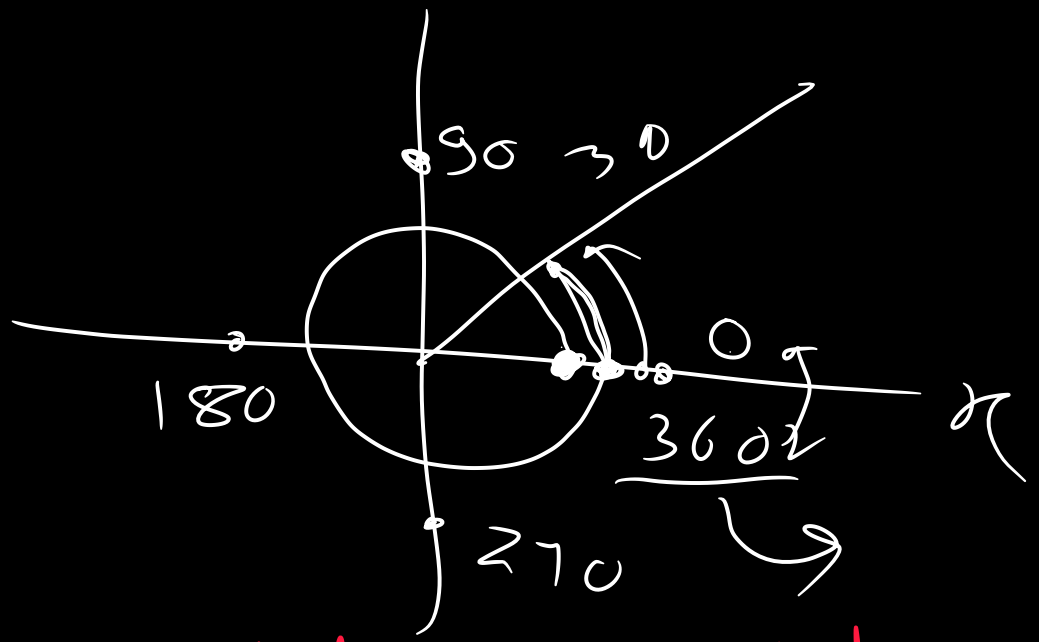
1. A → PT, B → QT, C → QT, D → PS
2. A → PT, B → QS, C → QT, D → PS
3. A → QT, B → QS, C → PT, D → PS
4. A → QS, B → PT, C → QT, D → PS




**Option 1 A → PT, B → QT, C → QT, D → PS**

|√3| > 1 so D is P and S, not T. |1/2| < 1 so A, B, C take T.

# Rough work / extra work




$360 + 30 = 390$

$\sin(0) = \sin(360)$

$\cos(0) = \cos(360)$

$\left(\frac{360}{2} + 30 = 390\right.$

As 1 rotation complete and we have $\theta = 360$

next value will → 1 not 361

After 360 ⇒ 1

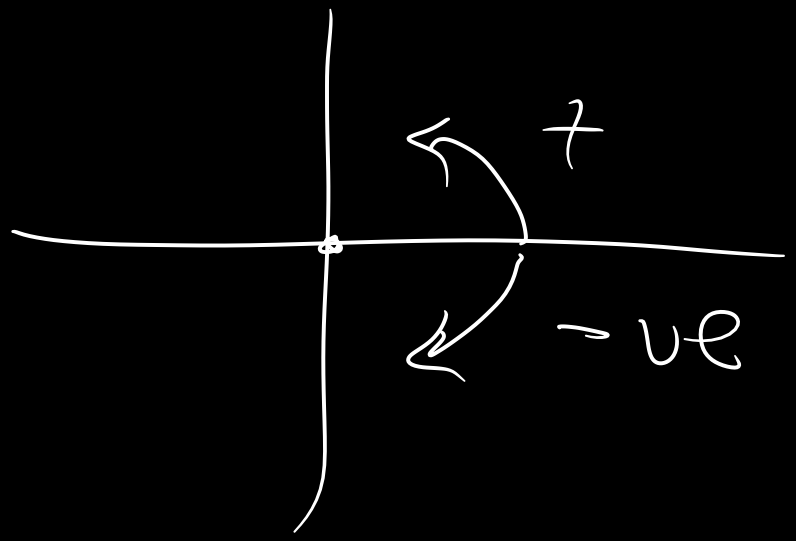
or 360 = 0

# Rough work / extra work

$390 = 360 + 30 = 30°$

$\sin(390) = \sin(30°) = \frac{1}{2}$

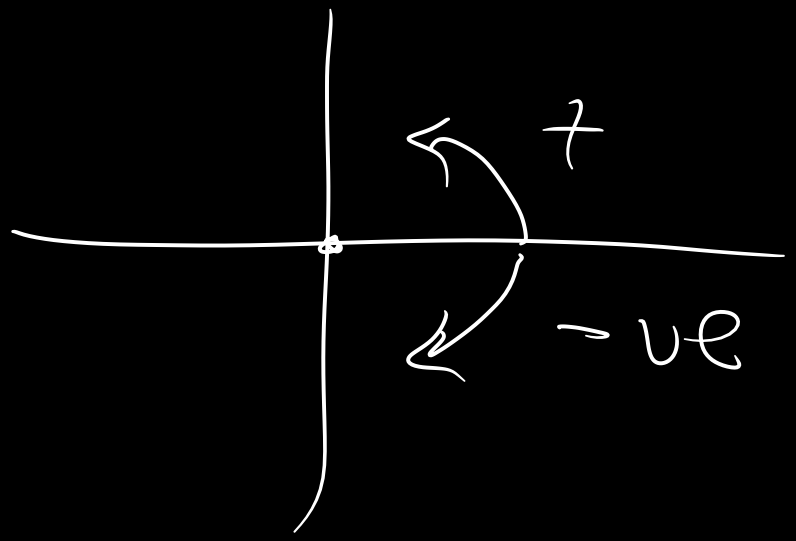
# Rough work / extra work




move in anticlockwise dirn → Angle will be in positive

If we move in clockwise direction angle will be negtive

# Rough work / extra work




$$\sin(-30) = -\frac{1}{2}$$

$$\cos(120) \Rightarrow \frac{-1}{2} = -\cos(90+\theta) = -\sin\theta$$

$$= -\sin(30)$$

$$= -\left(\frac{1}{2}\right) = -\frac{1}{2}$$

# Last class, slide 78 — after that class

That class asked you to write the six compound-angle lines ten times.
The two allied lines and the match-column are now answered on the board. Copy them once more.

**sin (180° − θ) = sin θ** **cos (180° + θ) = − cos θ**

**Match: A → PT, B → QT, C → QT, D → PS**

# Radian and arc — now locked

**180° = π rad**

Wherever you see π rad, put 180°.

degree → radian: multiply by π/180
radian → degree: multiply by 180/π
Arc length = R θ with θ in radian. Never R × 90°.

90° on a circle of radius R → distance πR/2

short notes

# Compound and prefixes — now locked

**sin (A + B) = sin A cos B + cos A sin B**

**cos (A + B) = cos A cos B − sin A sin B**

**tan (A + B) = (tan A + tan B) / (1 − tan A tan B)**

**fm femtometre $10^{-15}$ m** **nm nanometre $10^{-9}$ m** **Å ångström $10^{-10}$ m**

Do not write sin(A+B) = sin A + sin B. Nuclear size is fm, not mm.



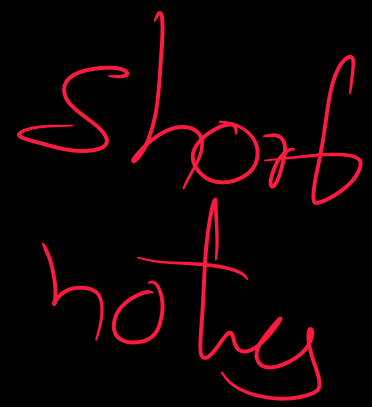
# Small angle — the one lock-line

$$\sin\theta \approx \theta \qquad \tan\theta \approx \theta \qquad \cos\theta \approx 1 \qquad (\theta \text{ in radian},\ \theta \le 5^\circ)$$

**$\sin 2^\circ \approx \pi/90$** **$\tan 3^\circ \approx \pi/60$** **$\cos 3^\circ \approx 1$**

Never leave the degree number in the approximation. Convert, then drop the word radian.

Leftover from last class is done. After the break: ASTC method, max–min, phase.



# BREAK

**Leftover from last class is done.  Next: ASTC method, max–min, phase**

# Three steps for any allied angle

θ is acute. The three steps write the value without opening the compound line every time.

1. **Decide the quadrant. Write the sign of that ratio from ASTC.**
2. **At $90° \pm \theta$ and $270° \pm \theta$ the function changes. At $180° \pm \theta$ and $360° \pm \theta$ it stays.**
3. **Change pairs: sin ↔ cos tan ↔ cot sec ↔ cosec**

**A S T C → All sin tan cos** → lecture 2

Do not look at the sign of the result after the change. Write the ASTC sign first, then the brother.

# Where each allied angle sits

| | | | | | |
|---|---|---|---|---|---|
| I | 0° to 90° | All +ve | 90° − θ | 360° + θ | |
| II | 90° to 180° | sin, cosec +ve | 90° + θ | 180° − θ | |
| III | 180° to 270° | tan, cot +ve | 180° + θ | 270° − θ | |
| IV | 270° to 360° | cos, sec +ve | 270° + θ | 360° − θ | −θ |

# Rough work / extra work

# Allied lines — 90° and 180°

$\sin(90° + \theta) = +\cos\theta$ $\quad$ $\sin(90° - \theta) = +\cos\theta$

$\cos(90° + \theta) = -\sin\theta$ $\quad$ $\cos(90° - \theta) = +\sin\theta$

$\tan(90° - \theta) = +\cot\theta$

$\sin(180° + \theta) = -\sin\theta$ $\quad$ $\sin(180° - \theta) = +\sin\theta$

$\cos(180° + \theta) = -\cos\theta$ $\quad$ $\tan(180° - \theta) = -\tan\theta$

$\cos(90° + \theta)$ is minus sine. That minus is the ASTC sign of cosine in the second quadrant.

# Allied lines — 270°, 360°, 450°, 540°

$\sin(270° + \theta) = -\cos\theta$ $\quad$ $\sin(270° - \theta) = -\cos\theta$

$\cos(270° + \theta) = +\sin\theta$ $\quad$ $\tan(360° + \theta) = +\tan\theta$

$\cos(360° - \theta) = +\cos\theta$ $\quad$ $\cos(-\theta) = +\cos\theta$

$\sin(450° + \theta) = \sin(90° + \theta) = +\cos\theta$

$\tan(540° + \theta) = \tan(180° + \theta) = +\tan\theta$

Drop full turns of 360° first. 450° = 90°. 540° = 180°. 720° = 0°.

# sin 300° and cos 240°

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**Find (1) sin 300° and (2) cos 240°. Two routes each.**

$\sin 300° = \sin(270° + 30°) = -\cos 30° = -\sqrt{3}/2$

$= \sin(360° - 60°) = -\sin 60° = -\sqrt{3}/2$

**(1) $\sin 300° = -\sqrt{3}/2$**

$\cos 240° = \cos(180° + 60°) = -\cos 60° = -1/2$

$= \cos(270° - 30°) = -\sin 30° = -1/2$

**(2) $\cos 240° = -1/2$**

# Past 360° — drop full turns first

**Find $\sin 405°$, $\cos 480°$ and $\cos 780°$.**

$\sin 405° = \sin(360° + 45°) = \sin 45° = 1/\sqrt{2}$

$\cos 480° = \cos(450° + 30°) = \cos(90° + 30°) = -\sin 30° = -1/2$

$\cos 780° = \cos(720° + 60°) = \cos 60° = 1/2$

$\sin 405° = 1/\sqrt{2}$ $\quad$ $\cos 480° = -1/2$ $\quad$ $\cos 780° = 1/2$

720° is two full turns. Cross it out. Do not open a compound line on 780°.

# Ten values from ASTC — read down

**Find each value. Drop 360° first. Then the three steps.**

| | angle | reduction | value |
|---|---|---|---|
| **(a)** | sin 30° | table | 1/2 |
| **(b)** | sin 45° | table | 1/√2 |
| **(c)** | sin 510° | sin (360° + 150°) = sin 150° | 1/2 |
| **(d)** | sin 240° | sin (180° + 60°) | −√3/2 |
| **(e)** | sin 330° | sin (360° − 30°) | −1/2 |
| **(f)** | cos 300° | cos (360° − 60°) | 1/2 |
| **(g)** | tan 135° | tan (180° − 45°) | −1 |
| **(h)** | cot 210° | cot (180° + 30°) = cot 30° | √3 |
| **(i)** | cos 510° | cos 150° = − cos 30° | −√3/2 |
| **(j)** | sin 135° | sin (180° − 45°) | 1/√2 |

# Rough work / extra work

# sin 74° from 2 × 37°

Find sin 74°. Last class: sin 37° = 3/5, cos 37° = 4/5.

sin 74° = sin (2 × 37°) = 2 sin 37° cos 37°

= 2 × (3/5) × (4/5) = 24/25

sin 74° = 24/25

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$\sin(74°) = \sin(2 \times 37°) = ?$

$= 2 \times \sin 37 \times \cos 37$

$= 2 \times \frac{3}{5} \times \frac{4}{5} \Rightarrow \frac{24}{35}$

# sin 22.5° from the half-angle line

H.W

**Find sin 22.5°. Last class: sin θ = √[(1 − cos 2θ) / 2].**

sin 22.5° = √[ (1 − cos 45°) / 2 ] = √[ (1 − 1/√2) / 2 ]

= √[ (√2 − 1) / (2√2) ]

sin 22.5° = [illegible]

Same line gives cos 15° = (√6 + √2)/4 and cos 105° = (√2 − √6)/4.

# BREAK

**After the break:  maximum and minimum,  then phase and phasor**

# Maximum and minimum of sine and cosine

$(\sin\theta)_{max} = +1$ at $\theta = 90°$

$(\cos\theta)_{max} = +1$ at $\theta = 0°$

$(\sin\theta)_{min} = -1$ at $\theta = 270°$

$(\cos\theta)_{min} = -1$ at $\theta = 180°$

$$-1 \le \sin\theta \le 1$$

$$-1 \le \cos\theta \le 1$$

A constant in front of sine is not the angle. $y = 6\sin\theta$ has ymax = 6, not $6 \times 90°$.

$\max(\sin\theta) = 1 \Rightarrow 90$

$\max(\cos\theta) = 1 \Rightarrow 360/0$

$\min(\sin\theta) \Rightarrow -1 \longrightarrow 270$

$\min(\cos\theta) = -1 \Rightarrow 180$

# $y = 6 \sin \theta$

**If $y = 6 \sin \theta$, find the maximum and minimum value of y.**

$y_{max} = 6 \times (\sin \theta)_{max} = 6 \times 1 = +6$ at $\theta = 90°$

$y_{min} = 6 \times (\sin \theta)_{min} = 6 \times (-1) = -6$ at $\theta = 270°$

$y_{max} = +6$ $\quad$ $y_{min} = -6$

$\theta = ? = 90$

$\frac{\pi}{2}$

$(6 \sin\theta)_{max} \Rightarrow max(6) \times max(\sin\theta)$

$(6) \times 1$ $\quad$ $-6 \rightarrow 90°$

$y_{min} = min(6 \sin\theta) = (6 \times -1) = -6$

$= 270°$

# The 3θ is the angle, not the 4

**If y = 4 sin (3θ), find ymax and the θ at which y is maximum.**

**(sin 3θ)max = +1 so ymax = 4 × 1 = +4**

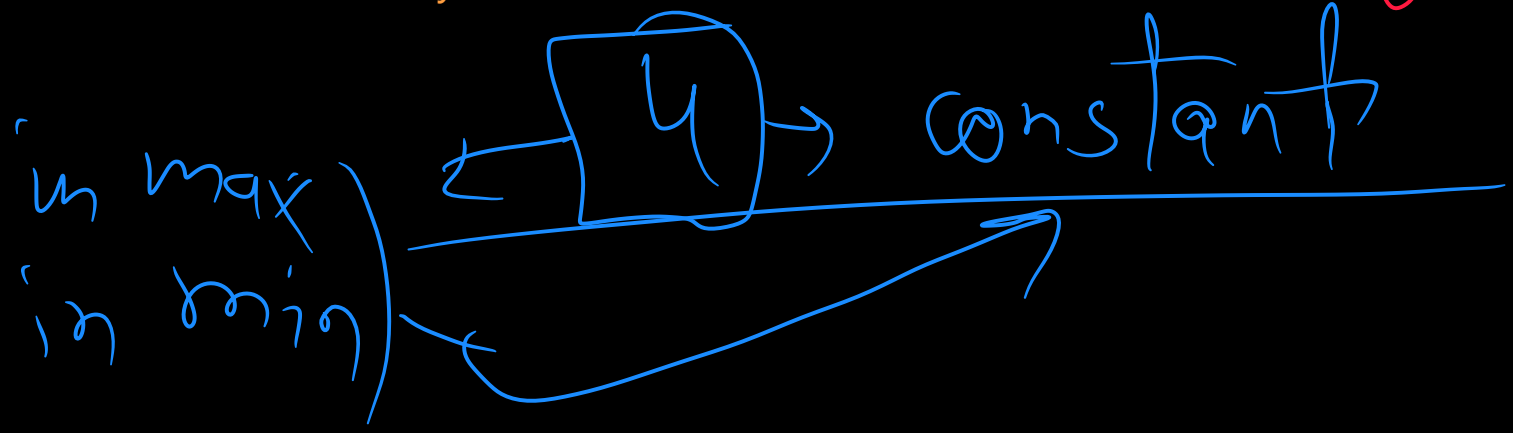
**The angle that is 90° is 3θ, not θ. 3θ = 90° θ = 30°**

**ymax = +4 at θ = 30°**

Do not set θ = 90°. The sine is of 3θ. Divide 90° by 3.

$y = 4 \sin(3\theta)$

$3\theta \Rightarrow$ Angle



# The 3θ is the angle, not the 4

**If y = 4 sin (3θ), find ymax and the θ at which y is maximum.**

**(sin 3θ)max = +1** **so** **ymax = 4 × 1 = +4**

**The angle that is 90° is 3θ, not θ.** **3θ = 90°** **θ = 30°**

**ymax = +4 at θ = 30°**

Do not set θ = 90°. The sine is of 3θ. Divide 90° by 3.

max (sin (3θ)) → 1

At θ = ?

sin (3θ°) = 1

3θ = 90 ⇒ θ = 90/3 = 30°

sin (100 × θ)

θ = 60

sin (6000)

360 = 0